

5-DAY HANDS-ON WORKSHOP FOR CLASS XII SCIENCE STUDENTS

FROM

CELLS

TO

CIRCUITS

A SMART HEALTH SYSTEMS JOURNEY



DATES

7 JULY – 11 JULY 2026

(TUESDAY – SATURDAY)



TIME

10:00 AM

TO 1:00 PM



VENUE

Shobhit University

Campus, Meerut



CERTIFICATE

Will be provided to

all participants



HANDS-ON
EXPERIMENTS &
LIVE DEMOS



EXPERT
FACULTY
MENTORSHIP



CERTIFICATE
OF PARTICIPATION



CAMPUS TOUR &
LAB VISITS



INNOVATION
CHALLENGE



GOODIES &
SURPRISES

5-DAY WORKSHOP SCHEDULE

DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
7 JULY 2026	8 JULY 2026	9 JULY 2026	10 JULY 2026	11 JULY 2026
BIOLOGICAL FOUNDATIONS	MICROSCOPY & CELL BIOLOGY	BIO SIGNALS & HUMAN PHYSIOLOGY	CIRCUITS BEHIND HEALTHCARE	INNOVATION CHALLENGE
 • Introduction to Biomedical Engineering • Biological Sciences • Human Physiology • Blood Components • Blood Grouping • Laboratory Safety	 • Blood Smear Preparation • Blood Staining • Microscopy • Cell Morphology (RBC, WBC, Platelets) • Exploring Animal Cell Culture Laboratory	 • Overview of Biosignals • Introduction to Signal Acquisition Systems • Overview of the Demo • Live Demo: ECG/EMG Acquisition • Real-time Signal Visualization	 • Group Activity / Circuit Connections • Module Amplifier Wiring & Pin Mapping • Interfacing Sensor Modules with Microcontroller • Serial Plotter / MATLAB / Python Visualization	 • Mini Project • Innovation Challenge • Prototype Presentation • Career Counselling • Certificate Distribution • Campus Tour
HANDS-ON ACTIVITY Blood Group Detection	HANDS-ON ACTIVITIES Blood Staining, Microscopy & Lab Visit	LIVE DEMO ECG/EMG Acquisition	HANDS-ON ACTIVITY Build & Visualize Your Own Signal System	GRAND FINALE Innovation, Guidance & Certificates

WORKSHOP HIGHLIGHTS

- ✓ Blood Grouping & Microscopy
- ✓ Cell Culture Lab Visit
- ✓ ECG & EMG Live Recording
- ✓ Biomedical Sensors & Electronics
- ✓ Arduino Programming
- ✓ Signal Visualization (MATLAB / Python)
- ✓ AI in Healthcare
- ✓ Innovation Challenge
- ✓ Certificate, Goodies & More!

EXPLORE THE FUTURE OF BIOMEDICAL ENGINEERING



CAREER PATHS

- Clinical Engineer
- Medical Device Engineer
- Healthcare AI Engineer
- Biomedical Researcher
- Health Data Scientist
- Medical Robotics Engineer
- Rehabilitation Engineer
- Biomedical Startup Founder



LIMITED SEATS!

REGISTER NOW

Be part of the journey from

Biology to Technology to Innovation!

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Workshop Website

www.shobhituniversity.ac.in